

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Association of Biliary Complications and Treatment Duration in Unresectable BTC Patients
Short Title/Acronym: ARCBile **Protocol Number:** D4191R00099 **NCT Number:** NCT07101874 **Sponsor Name:** AstraZeneca (in collaboration with Japan Oncology Network in Hepatobiliary and Pancreas - JON-HBP) **Investigational Product:** First-line systemic therapy for unresectable BTC (Routine clinical care) **Phase of Development:** Observational **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the association of the occurrence of biliary complications requiring hospitalization during first-line systemic therapy with time to treatment failure (TTF) in unresectable Biliary Tract Cancer (BTC) patients. **Secondary Objectives:** To evaluate the real-world evidence on cancer-related biliary complications, assess treatment duration, and clarify priority issues for managing advanced BTC.

2.2 Background & Rationale The real-world evidence on cancer-related biliary complications during systemic therapy in patients with advanced biliary tract cancers (BTC) in a large population is lacking. The results from this study attempt to test the hypothesis that the occurrence of biliary complications requiring hospitalization as a time-dependent covariate significantly affects TTF. It aims to reveal the association of biliary complications with treatment duration, provide an overview of the current status of managing them, and clarify the issues to be addressed as a priority in future studies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective, multi-center) **Control Type:** N/A (Observational, real-world data) **Blinding:** Open-label / N/A (Observational) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 1,000 patients **Number of Sites:** 22 locations in Japan **Estimated Study Start:** 22-Aug-2025 **Estimated Primary Completion:** 28-Feb-2026 **Estimated Study Completion:** 28-Feb-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients aged 18 years or older (All Genders) at the date of first dose of first-line systemic therapy. 2. Patients diagnosed with "unresectable BTC" (including intrahepatic/perihilar/distal cholangiocarcinoma, GBC, or cancer of the AoV) who meet any of the following: deemed unamenable to curative resection at initial diagnosis, or relapsed after curative resection of BTC > 6 months after the end of adjuvant therapy or surgery for patients who did not receive adjuvant therapy. 3. Patients who received first-line systemic therapy for unresectable BTC with the date of the first dose between May 1, 2022, and December 31, 2023. 4. Patients with ECOG performance status 0 or 1 on the most recent assessment date prior to first dose. 5. Patients who started first-line systemic therapy with physician's expectation of 3 months or longer prognosis at the time of starting systemic therapy.

4.2 Exclusion Criteria 1. Patients who received postoperative adjuvant therapy for BTC with R2 resection. 2. Maintenance dialysis at the date of the first dose of first-line systemic therapy for unresectable BTC. 3. Active cancers other than unresectable BTC within the past 2 years from the date of the first dose of first-line systemic therapy. Exceptions: Carcinoma in situ or lesions equivalent to carcinoma in situ; intramucosal cancers that are deemed cured by local treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: As prescribed (First-line systemic therapy as routine clinical care). **Route of Administration:** Intravenous / Oral (Standard systemic therapy regimens). **Duration of Treatment:** Data evaluated from 1st dose of first-line systemic therapy until disease progression, death by any cause, or discontinuation by physician's decision, assessed up to 2 years (data cut-off July 31, 2025).

5.2 Concomitant & Prohibited Therapy Permitted: Standard routine clinical care medications. **Prohibited:** N/A (Observational study).

6. Study Timeline & Visit Schedule

(Note: As a retrospective observational study, there are no prospective patient visits. Schedule denotes the timeline of data abstraction.) | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Historical | Patient records evaluation for inclusion (1st dose between May 2022 and Dec 2023) | | **Baseline** | Day 0 | Index date defined as the start of first-line systemic therapy | | **Interim** | N/A | Longitudinal abstraction of medical charts for biliary complication occurrences | | **End of Study** | Up to 2 years |

Follow-up data collection until Jul 31, 2025, or occurrence of treatment failure/death |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The association of biliary complications requiring hospitalization during first-line systemic therapy with time to treatment failure (TTF).

Measurement Method: Evaluated by Hazard Ratio (HR) with a 95% confidence interval (CI), estimated with TTF and the incidence rate of the event in patient groups who developed biliary complications before a certain time t versus those who did not, incorporating it as a time-dependent covariate.

7.2 Secondary & Exploratory Endpoints Incidence rate of biliary complications requiring hospitalization during the treatment window. Treatment duration insights regarding managing complications in advanced BTC.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective medical chart review focusing strictly on the occurrence of biliary complications requiring hospitalization.

Serious Adverse Events (SAEs):

Evaluated strictly via historical hospitalization records mapped to standard clinical care.

Data Monitoring Committee (DMC):

Managed centrally via the Japan Oncology Network in Hepatobiliary and Pancreas (JON-HBP) data registry.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible patients who initiated first-line systemic therapy for unresectable BTC between May 2022 and December 2023.

Sample Size Justification: Approximately $N=1,000$ patients to provide adequate statistical power to analyze time-dependent covariates for biliary complications over a 2-year window.

Handling of Missing Data: Standard retrospective rules for observational studies, accounting for censoring if patients are lost to follow-up prior to July 2025.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Epidemiological Practice (GEP) guidelines and local observational study regulations.

Monitoring Plan: Retrospective source data abstraction and verification from JON-HBP electronic medical records.

Audit Trail: Centralized database registry tracking abstraction parameters and querying for missing chart elements.

11. Study Identifiers & Governance

Primary ID: D4191R00099 **Registry Number:** NCT07101874 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARCBile **Official Regulatory Title:** A Retrospective Study of Biliary Complications and Treatment Duration in Patients With Biliary Tract Cancer Receiving Systemic Therapy in Japan

12. Clinical Framework (Biliary Tract Cancer Specific)

Primary Condition: Biliary Tract Cancer (BTC) **Diagnosis Threshold:** Unresectable BTC (including intrahepatic/perihilar/distal cholangiocarcinoma, GBC, or cancer of the AoV) deemed unamenable to curative resection or relapsed >6 months post-adjuvant therapy. **Medical Methodology:** First-line Systemic Therapy **Optimization Target:** Minimization of time to treatment failure (TTF) and mapping of biliary complications

13. Study Procedures & Methodology

Management Pathway: Routine clinical care systemic therapy pathway for advanced BTC in Japan. **Education Protocol (HCP):** N/A (Retrospective chart review) **Education Protocol (Patient):** N/A (Retrospective chart review) **Quality Audit Mechanism:** Abstracted data validation through JON-HBP network records.

14. Observation & Follow-Up Schedule

Total Duration: Retrospective observation window up to 2 years (Index dose to July 31, 2025). **Onsite Visit Cadence:** N/A (Retrospective database extraction) **Remote Monitoring Frequency:** Continuous historical abstraction. **Checklist Review Interval:** Final cohort analysis generated post-abstraction (primary completion Feb 2026).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				